Sharp Defect Thresholds for Direct Reconfiguration of Six Perfect Matchings

Research draft — mathematical verification complete; novelty not externally established

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Abstract

For six perfect matchings of a simple cubic bipartite graph, let Q be the sum of squared deviations of edge multiplicities from two, and let κ be the minimum number of matchings whose simultaneous replacement produces an exact double cover. We prove that the smallest defects at which $\kappa \geq 2, 3, 4$ are respectively $6, 12, 34$. In particular, $Q \leq 10$ forces a direct two-coordinate exact repair, while $Q \leq 32$ forces a direct three-coordinate exact repair. The new upper threshold is attained by a certified 32-vertex 3-edge-connected cubic bipartite graph with $Q = 34$ and $\kappa = 4$. The proofs combine bipartite factorization, an exact multiplicity energy identity, and a finite covering argument on the twenty triples of six matching labels. This concerns *one-step exact repair*, not arbitrary strict two-coordinate descent or a solution of Berge–Fulkerson.

1 Definitions and the completion criterion

Let $G = (U \sqcup W, E)$ be a finite simple cubic bipartite graph. Take a sextuple $\mathcal{M} = (M_0, \ldots, M_5)$ of perfect matchings, with repetitions allowed. For each edge e, define $S_e = \{i : e \in M_i\}$, $m_e = |S_e|$, and

$$Q(\mathcal{M}) = \sum_{e \in E} (m_e - 2)^2.$$

Let $\kappa(\mathcal{M})$ be the minimum cardinality of a set $I \subseteq \{0, \ldots, 5\}$ of matching positions that can be replaced, leaving the other positions frozen, in such a way that every edge finally has multiplicity exactly two. By convention, $\kappa = 0$ if $Q = 0$.

Lemma 1 (Exact frozen-factor criterion)**.** *For $I \subseteq [6] = \{0, \ldots, 5\}$, a direct repair of the coordinates in I exists if and only if*

$$|I \cap S_e| \geq \max(0, m_e - 2) \quad \text{for every } e \in E. \tag{1}$$

In particular, $\kappa \leq 4$ for every sextuple.

Proof. Necessity holds because no frozen edge can remain covered more than twice. For sufficiency, the residual edge demand after freezing the positions outside I is $r(e) = 2 - m_e + |I \cap S_e| \geq 0$. At each vertex the sum of these residual demands is $6 - (6 - |I|) = |I|$. Thus r is an integral $|I|$-regular bipartite multigraph on the same vertex set. Repeated application of the bipartite perfect-matching theorem decomposes r into $|I|$ perfect matchings. For $|I| = 4$, at most two coordinates are frozen, so every frozen multiplicity is at most two. □

The identity $\sum_e (m_e - 2) = 0$ gives

$$\frac{Q}{2} = \sum_{e \in E} \frac{(m_e - 2)(m_e - 1)}{2} = n_0 + n_3 + 3n_4 + 6n_5 + 10n_6, \tag{1}$$

where n_j is the number of edges covered by exactly j matchings.

Theorem 2 (Universal two-coordinate threshold)**.** *If $0 < Q \leq 10$, then $\kappa \leq 2$. The bound is sharp: an explicit simple 3-edge-connected cubic bipartite graph has $Q = 12$ and $\kappa = 3$.*

Proof. Equation (1) excludes multiplicities five and six when $Q \leq 10$ and permits at most one edge of multiplicity four. If no edge has multiplicity four, the surplus membership sets S_e are at most five triples of [6]. There exists a pair $I \subseteq [6]$ meeting every one of them. Indeed, otherwise the complementary triples $[6] \setminus S_e$ would cover all fifteen pairs of [6]. Five triangles covering all fifteen edges of K_6 would form a triangle decomposition of K_6, impossible because every vertex of K_6 has odd degree. If there are fewer than five triples, counting alone suffices.

If one surplus edge has multiplicity four, (1) allows at most two surplus triples. Choose a pair I contained in that four-element membership set F. Each additional triple T forbids at most the three pairs contained in $F \setminus T$, which has size at most three. Two triangles cannot partition the six edges of K_4 (distinct triangles share an edge). Therefore a pair $I \subseteq F$ meets both triples. In either case (1) holds and Lemma 1 proves direct repair. The sharp $Q = 12$ witness, with an independent checker, is recorded in the accompanying earlier research branch. □

2 The first possible four-coordinate obstruction

For a three-element set $I \subseteq [6]$, call I *blocked* at a vertex $v \in U$ if some edge e incident with v violates (1). Every edge has exactly one endpoint in U. Hence $\kappa \geq 4$ if and only if the blocked sets of the vertices of U cover all 20 triples I.

The three edge-label sets incident at any vertex partition [6]. Up to order, the seven possible local multiplicity triples, their costs $\sum(m-2)^2$, and the number of triples they block are:

Local multiplicities	Cost	Blocked triples
$(2,2,2)$	0	0
$(3,2,1)$	2	1
$(3,3,0)$	6	2
$(4,1,1)$	6	4
$(4,2,0)$	8	4
$(5,1,0)$	14	10
$(6,0,0)$	24	20

An $(4,1,1)$ or $(4,2,0)$ vertex blocks the four triples containing the two labels absent from its four-fold edge. A $(5,1,0)$ vertex blocks the ten triples containing the one label absent from its five-fold edge. These explicit forms are used below.

Lemma 3 (Twenty-triple covering inequality)**.** *Suppose a multiset of partitions of* [6] *into three parts contains no* $(6,0,0)$ *type, and its blocked triples cover all* 20 *triples of* [6]*. Then the sum of its local costs is at least* 34*.*

Proof. Suppose the cost were at most 32. Let a count vertices of type $(3,2,1)$; let b count the types $(3,3,0)$, $(4,1,1)$, and $(4,2,0)$; and let c count the type $(5,1,0)$. Neutral vertices do not matter. Then

$$2a + 6b + 14c \leq 32, \qquad 20 \leq a + 4b + 10c. \tag{2}$$

The second inequality is a generous upper bound: some b-types block only two triples and some cost eight.

If $c = 0$, the inequalities force $b \geq 4$. For $b = 4$, they force $a \leq 4$; coverage of all twenty triples would require four pair-blocks of size four to be mutually disjoint. Their corresponding two-element label sets would have to be four mutually disjoint pairs on six labels, impossible. For $b = 5$, at most one extra singleton block is available. The five pair-blocks omit at least two triples: equivalently, every five-edge graph on six vertices has at least two independent vertex triples. To verify this elementary claim, its complement has ten edges. Mantel's theorem gives a triangle; if this were its only triangle, deleting an edge of that triangle would give a nine-edge triangle-free $K_{3,3}$, whereas adding a single edge to $K_{3,3}$ creates three triangles. Thus at least two triangles exist in the complement. The case $b \geq 6$ exceeds the budget.

If $c = 1$, the inequalities give $b \leq 3$ and $a \leq 9 - 3b$, so the crude covering capacity is at most $19 + b$. Thus $b \geq 1$. A four-block and a ten-block always intersect: a two-label pair and a single label are contained together in at least one triple. For $b = 1$ there are only twenty possible block-slots, so this overlap prevents coverage. For $b = 2$, there are at most twenty-one slots, but two distinct four-blocks each intersect the ten-block in a distinct triple (unless one intersects it in four triples); repeated four-blocks only waste more slots. Thus at least two slots are duplicated. For $b = 3$, there are at most twenty-two slots and at least three duplicated slots from the three distinct four-blocks meeting the ten-block; repeated blocks also waste at least three. A b-type with only two blocks loses two slots and cannot improve any case. Hence no b works.

If $c = 2$, then $b = 0$ and $a \leq 2$. Two ten-blocks cover at most sixteen of the twenty triples (those containing at least one of the two omitted labels), leaving at least four uncovered. Finally $c \geq 3$ exceeds the cost budget. This contradiction proves the lemma. □

Lemma 4 (Sixfold-edge energy barrier)**.** *If G is simple cubic and some edge has multiplicity six, then $Q \geq 36$.*

Proof. That edge contributes 16 to Q. The other four edges incident with its endpoints are distinct by simplicity and have multiplicity zero, contributing another 16. The deviations on these five edges sum to $4 - 4(2) = -4$. Since total deviation is zero, the remaining edges have deviations summing to 4 and squared deviations totaling at least 4. Thus $Q \geq 36$. □

Theorem 5 (Sharp unrestricted triple-repair bound)**.** *Every sextuple of perfect matchings in a simple cubic bipartite graph with $Q \leq 32$ admits an exact repair of at most three matching coordinates. This is sharp: there exists a simple 3-edge-connected cubic bipartite graph with 32 vertices, six perfect matchings, $Q = 34$ and $\kappa = 4$.*

Proof. If $\kappa \geq 4$, the vertex blocking sets on U cover all twenty triples. Since each edge is counted once at its endpoint in U, Q is exactly the sum of the local costs over U. If no edge has multiplicity six, Lemma 3 gives $Q \geq 34$; otherwise Lemma 4 gives $Q \geq 36$. Thus $Q \leq 32$ implies $\kappa \leq 3$ by Lemma 1.

For sharpness, use the explicit edge-label table in Appendix A. It defines a cubic bipartite graph on $u_0, \ldots, u_{15}$ and $v_0, \ldots, v_{15}$: an entry $(j; S)$ denotes the edge $u_i v_j$ belonging to exactly the matching colours in S. At every vertex the three incident label sets partition $[6]$, so each colour forms a perfect matching. Direct counting gives $n_1 = 14$, $n_2 = 23$, $n_3 = 8$, $n_4 = 3$ and $Q = 14 + 8 + 3(4) = 34$.

The three fourfold-edge label sets are 2345, 0145, 0123. Their complements are the three disjoint pairs 01, 23, 45. A choice I of three matching positions containing any one of these pairs is blocked. The eight remaining triples I each select exactly one label from each pair; they are the complements of the eight distinct threefold-edge sets $024, 025, 034, 035, 124, 125, 134, 135$. Each remaining I is consequently blocked too. Thus $\kappa \geq 4$; Lemma 1 gives $\kappa \leq 4$. Finally, the accompanying standalone verifier checks connectedness after deleting any set of at most two edges, proving 3-edge-connectivity. The finite verification is small enough to reproduce independently without an optimization solver. □

Lemma 6 (Smallest positive-defect repair)**.** *If G is simple cubic bipartite and $0 < Q < 6$, then $Q = 4$ and $\kappa = 1$.*

Proof. Any edge with multiplicity zero forces, at either endpoint, two incident edges of multiplicity three; by simplicity the four incident threefold edges are distinct. Thus (1) gives $Q/2 \geq 1 + 4 > 2$. An edge of multiplicity four or more also forces $Q/2 \geq 3$. Hence $Q < 6$ allows only unit deviations, and the support of nonzero deviations is an alternating union of even cycles: at a defective vertex its incident multiplicities are $(1, 2, 3)$. A nonempty such support in a simple graph has at least four edges. Thus $Q = 4$ and the support is one alternating four-cycle. Its two surplus edges have label triples T_1, T_2. A singleton deficit edge is incident with each of the two surplus edges, so its matching label belongs to neither T_1 nor T_2. Consequently $T_1 \cup T_2$ has fewer than six elements, which forces $T_1 \cap T_2 \neq \varnothing$. The perfect matching of a label in this intersection uses both surplus edges and neither deficit edge; flipping its edges on the four-cycle repairs the complete defect. Hence $\kappa = 1$. □

Corollary 7 (Full hierarchy)**.** *Within finite simple cubic bipartite graphs, even restricting to 3-edge-connected graphs, the least possible positive defects for $\kappa \geq 2$, $\kappa \geq 3$, and $\kappa \geq 4$ are respectively*

$$\boxed{6, \quad 12, \quad 34}.$$

Proof. For $\kappa \geq 2$, the established alternating-four-cycle repair lemma and the prior explicit $Q = 6$ witness give the threshold six. Theorem 2 and the prior explicit $Q = 12$ witness yield twelve. Theorem 5 gives thirty-four. □

Remark 8 (Unit-defect qualification)**.** *For the subfamily where all edge multiplicities lie in $\{1, 2, 3\}$, the corresponding $\kappa \geq 4$ minimum remains $Q = 40$, witnessed by a prior 40-vertex graph. The new $Q = 34$ example uses three edges of multiplicity four. Neither result implies the existence of a strict two-coordinate local minimum: exact one-step repair and descent are different questions. Indeed the displayed $Q = 34$ certificate has a two-coordinate improvement from 34 to 8.*

3 Research status and reproducibility

Existing matching-reconfiguration work examines transformations of individual perfect matchings, often by short alternating-cycle flips; see Bonamy et al. [1]. Recent work of Mori et al. [2] interprets transformations of collections of matchings with the same aggregate incidence vector through toric ideals and edge colourings. Our frozen-coordinate question instead allows the aggregate multiplicity vector to change, but asks how many entire matching coordinates must be replaced *at once* to attain the distinguished aggregate $\mathbf{21}$. This distinction does not prove novelty; a broader specialist audit of covering-design and edge-colouring literature is required.

The result is a new candidate theorem within our research programme, supported by a complete human-readable proof and a solver-independent finite witness. It has not been assessed by an external graph theorist and its novelty relative to the complete matching reconfiguration and edge-colouring literature remains to be established. The classical ingredient in Lemma 1 is Kőnig's

bipartite factorization theorem. The 32-vertex JSON certificate and independent Python checker are under `berge_fulkerson/plateau/next_phase/` in the research branch. Separately, a standard-library C++17 finite cover enumerator (not needed by the proof) independently certifies that the minimum cost of twenty-triple coverage using partitions without a sixfold edge is 34.

Transparency statement. AI-assisted mathematical exploration and computational search helped formulate conjectures, discover extremal instances, and prepare this draft. The mathematical statements, proofs and data have been separated from heuristic outputs; the graph certificates can be verified without access to the discovery tools. Formal Lean verification and independent specialist refereeing remain outstanding.

References

[1] M. Bonamy, N. Bousquet, M. Heinrich, T. Ito, Y. Kobayashi, A. Mary, M. Mühlenthaler, and K. Wasa, *The Perfect Matching Reconfiguration Problem*, MFCS 2019, LIPIcs 138, article 80, DOI: 10.4230/LIPIcs.MFCS.2019.80.

[2] K. Mori, R. Motomura, H. Ohsugi, and A. Tsuchiya, *Toric ideals of matching polytopes and edge colorings*, Mathematika 72 (2026), DOI: 10.1112/mtk.70099.

A Explicit 32-vertex certified witness

In the following table, the symbols in a label-set are matching indices $0, \ldots, 5$. Every row gives the three neighbors of u_i, and each v_j is in the other bipartition class. The same data are archived in machine-readable JSON with a complete independent verifier.

i	$(j; S)$	$(j; S)$	$(j; S)$
0	$(0; \{2,3,4,5\})$	$(4; \{1\})$	$(10; \{0\})$
1	$(1; \{0,1,4,5\})$	$(8; \{3\})$	$(9; \{2\})$
2	$(2; \{0,1,2,3\})$	$(5; \{5\})$	$(6; \{4\})$
3	$(2; \{5\})$	$(3; \{0,2,4\})$	$(12; \{1,3\})$
4	$(0; \{1\})$	$(4; \{0,2,5\})$	$(14; \{3,4\})$
5	$(3; \{5\})$	$(5; \{0,3,4\})$	$(11; \{1,2\})$
6	$(2; \{4\})$	$(6; \{0,3,5\})$	$(15; \{1,2\})$
7	$(0; \{0\})$	$(7; \{1,2,4\})$	$(15; \{3,5\})$
8	$(1; \{3\})$	$(8; \{1,2,5\})$	$(15; \{0,4\})$
9	$(1; \{2\})$	$(9; \{1,3,4\})$	$(12; \{0,5\})$
10	$(7; \{0\})$	$(10; \{1,3,5\})$	$(12; \{2,4\})$
11	$(6; \{1,2\})$	$(11; \{4,5\})$	$(13; \{0,3\})$
12	$(3; \{1,3\})$	$(9; \{0,5\})$	$(10; \{2,4\})$
13	$(11; \{0,3\})$	$(13; \{1,4\})$	$(14; \{2,5\})$
14	$(4; \{3,4\})$	$(13; \{2,5\})$	$(14; \{0,1\})$
15	$(5; \{1,2\})$	$(7; \{3,5\})$	$(8; \{0,4\})$

Independent checks: 32 vertices, 48 distinct edges, 3 edges at every vertex, exactly one incident edge of every matching colour, $Q = 34$, $\kappa = 4$, and connectedness after all $1 + 48 + \binom{48}{2} = 1177$ deletions of zero, one, or two edges.